Preliminary Report:

ISYE 521: Machine Learning in Action- Authors: Mason Reitmeier, Dylan Hicks, Justin Zens

Problem statement:

This project investigates how accurately machine learning methods can predict apartment housing prices in the Moscow metropolitan area. Using a Kaggle dataset of over 22,000 listings with 11 features, we aim to answer three questions: (1) How accurately can we predict apartment prices given the available variables? (2) Which apartment features are most important in determining price? (3) Since price reflects buyer preference, what do Moscow residents appear to prioritize when searching for housing?

Data:

The dataset was sourced from Kaggle and originally contained 22,676 apartment listings in the Moscow metropolitan area with 11 features and one target variable (Price, in rubles). Features include continuous variables (Area, Living area, Kitchen area, Minutes to metro), discrete variables (Number of rooms, Floor, Number of floors), categorical variables (Apartment type, Renovation), a binary variable (Region: Moscow vs. Moscow region), and a high-cardinality nominal variable (Metro station, 317 unique stations).

Methods:

Data Cleaning: We removed 1,851 duplicate rows and 2,451 rows with invalid entries where floor exceeded the building's total floors, kitchen area exceeded total area, or the sum of kitchen and living area exceeded total area. No missing values were present in the raw data. The final cleaned dataset contains 18,374 observations across 12 columns. Median apartment price is approximately 12.4 million rubles, with a median area of 54.7 square meters and a median of 2 rooms, though the distribution has a long right tail with listings exceeding 2 billion rubles. For modeling, categorical variables (Apartment type, Region, Renovation) were one-hot encoded with the first category dropped, Metro station was excluded due to its high cardinality, and the target variable was log-transformed to address severe right skewness (raw skewness of 7.63 reduced to 1.13). The data was split 80/20 into training (14,699) and test (3,675) sets, yielding 12 input features.

Preprocessing: The target variable (Price) was log-transformed to address severe right skewness (raw skewness of 7.63 reduced to 1.13). Categorical variables (Apartment type, Region, Renovation) were one-hot encoded with the first category dropped to avoid multicollinearity. Metro station was excluded due to its high cardinality (317 unique values). All models were trained on an 80/20 train-test split (14,699 / 3,675 observations, random state = 42) and evaluated using R^2, RMSE on log-price, and MAE in rubles.

OLS Linear Regression (Baseline): A standard ordinary least squares regression was fitted using all 12 features to establish a performance baseline. This model assumes a linear relationship between features and log(Price) and provides interpretable coefficients for understanding feature importance.

Lasso Regression: To perform automatic feature selection and address potential multicollinearity (notably between Area and Living area, $r = 0.91$), we fit a Lasso regression using a pipeline that standardizes numeric features before regularization. The optimal regularization parameter (alpha) was selected via 10-fold cross-validation. Lasso penalizes the absolute magnitude of coefficients, shrinking less important features toward zero and potentially removing them entirely.

Random Forest: To capture the non-linear relationships of our features and to pinpoint the non-linear trends, we used a series of decision trees and a random forest model. We used one-hot encoding to split features for the initial decision tree. We then optimized the model and created a random forest through cross-validation. The goal of Random Forest was to reduce variance from the initial Decision Tree model.

Initial Findings:

EDA Findings:

The target variable, Price, is heavily right-skewed (skewness = 7.63) with a median of 12.4 million rubles and a maximum exceeding 2.4 billion rubles. A log transformation reduced skewness to 1.13 and produced a much more symmetric distribution suitable for linear modeling.

Correlation analysis against log(Price) revealed Area (r = 0.80) as the strongest numeric predictor, followed by Living area (0.74), Number of rooms (0.67), and Kitchen area (0.63). Floor (0.11), Number of floors (0.08), and Minutes to metro (-0.08) showed weak associations. Notably, Area and Living area are highly correlated with each other (r = 0.91), signaling potential multicollinearity in linear models. Price variance increases with apartment size, visible in scatter plots as a widening fan shape, indicating heteroscedasticity that the log transformation partially addresses.

Most numeric features are right skewed as well. Most apartments have 1-3 rooms, are under 100 square meters, and sit on lower-to-mid floors. These distributions suggest that much of the dataset represents typical mid-range housing, with a smaller tail of luxury properties driving the skew.

Among categorical features, apartments in Moscow command a clear premium over the Moscow region, and renovation quality shows a strong price gradient: Designer renovations sit at the top, followed by European-style, Cosmetic, and Without renovation at the bottom. Apartment type also shows meaningful price differences, with new buildings priced higher on average than Secondary (resale) apartments. The dataset is roughly balanced between Moscow (city) and Moscow region, while Secondary apartments outnumber new buildings.

Linear Regression Model:

OLS Results: The baseline OLS model explained approximately 87% of the variance in log(Price) on the test set, with nearly identical train and test performance, suggesting good generalization. Coefficient analysis revealed that categorical features like Designer renovation, Region (Moscow), and Without renovation (negative effect) had the largest absolute coefficients, while continuous features like Area exert smaller per-unit effects but remain important due to their range.

Lasso Results: Lasso regression with 10-fold CV selected an optimal alpha of 0.000852 and zeroed out one feature: Floor. This confirms the EDA finding that floor number is not a meaningful predictor once other features are accounted for. Despite the high collinearity between Area and Living area, Lasso retained both but heavily penalized Living area (coefficient of 0.02 vs. 0.40 for Area), effectively identifying Area as the dominant size predictor. Overall performance was virtually identical to OLS, indicating the original feature set was already lean with little noise to regularize away.

Metric	OLS (Baseline)	Lasso (CV)
Train R^2	0.8599	0.8599
Test R^2	0.8694	0.8694
RMSE (log)	0.3888	0.3889
Features used	12	11

Table 1: OLS Baseline and Lasso Regression Comparison

Random Forest Model:

Before creating a random forest model, we began this analysis with a basic Decision Tree model to establish a baseline for non-linear predictions. The initial tree was, out of the box, able to predict apartment prices with an R2 of 0.689. To improve this and prevent any overfitting with the deep, complex decision tree, we optimized the model using grid search cross-validation to tune the hyperparameters. After adjustment, the Optimized Decision Tree performed even better, with an R2 of 0.771. This initial tree indicated that tree-based models are good at capturing the relationships between apartment features and price, and that a Random Forest model was worth constructing.

Because of the initial Decision Tree success, we built a Random Forest Regression model. We chose Random Forest due to its strong capabilities in reducing variance and increasing model accuracy. We used a randomized search cross-validation to optimize the hyperparameters in the model. The Optimized Random Forest outputted our strongest results in the project, returning an R2 value of 0.825. This means that the model can successfully predict roughly 82.5% of apartment price variation with the given features. Compared to the Linear, AdaBoost, and single Decision Tree models, the Random Forest proved to be the most accurate approach for answering the key question of predicting housing prices. Additionally, by using the same testing and training data to identify the most important features in these models, it was again shown that Area is the most important predictor of Moscow apartment prices.

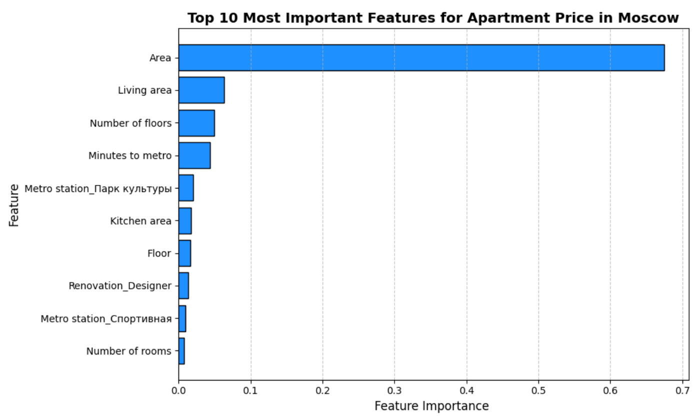

Figure 1: Most Important Features for Apartment Price

<u>Completion Plan:</u>

We have completed 3 of the 4 planned models (Linear, Random Forest, AdaBoost), so to finish the project, we will conduct a neural network model to see if a more involved approach is required to maximize prediction capabilities. The data is clean and works nicely with the models we have implemented so far. In this stage, we will continue tuning hyperparameters of each of the 3 models we have implemented so far to maximize R2 values.

To gain some real-world intuition and learning from this project, we will explore the root causes behind the most impactful features in the dataset. For example, we could easily predict that square footage would be positively associated with an increase in price since in general, larger apartments tend to be more expensive. However, there are some other confounding variables, such as Renovation level and style, that may give us further, unique insight into the preferences of Moscow apartment renters. We will continue exploring these possibilities for our final report using the findings of our completed models.